E-commerce User Behavior Analysis & Recommendation System

CS5530 Principles of Data Science ▪ University of Missouri–Kansas City ▪ Spring 2026
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Introduction

Background
E-commerce platforms generate millions of behavioral events daily, but raw event logs alone reveal very little information. Through data transformation and modeling, patterns in browsing, purchasing, and pricing behavior can be uncovered to drive decisions for both businesses and consumers. This project builds a seven-component pipeline on a publicly available dataset of 110 million time stamped events to address this challenge end-to-end.

Project Aims
- Analyze user browsing and purchasing patterns
- Build a collaborative filtering recommendation system
- Create user segmentation based on behavior
- Implement A/B testing across user groups
- Develop a time-sensitive recommendation layer
- Visualize the full customer journey including cart abandonment
- Predict optimal purchase timing based on price history

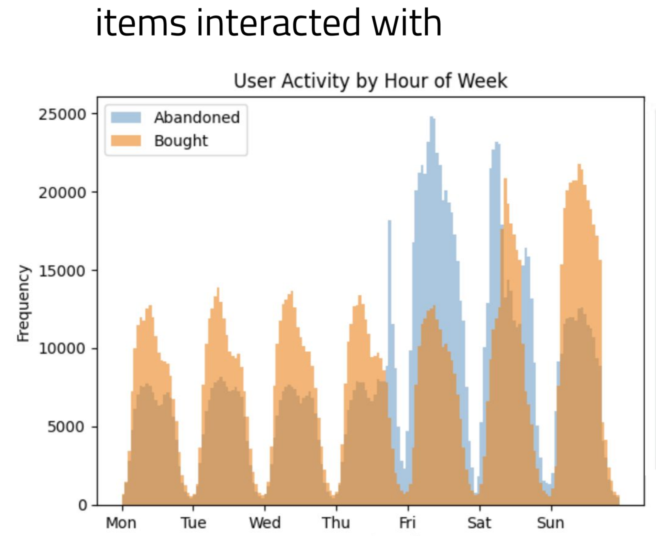
Key Results
- Four distinct user groups
- CF Recall@5: 62.39% for 9 of 13 categories (above 43.45% random baseline)
- Temporal re-ranking NDCG@5: 0.2120 vs 0.2117 baseline
- Price-timing ROC-AUC: 0.744 vs Recall: 0.832
- 150K products assigned to buy/wait signals

Customer Journey

Methods
- Visualized likelihood of all outcomes of user sessions.
- Used both original data set and session features data set
- Found patterns within items that were added to cart but not bought

```
Viewed: 12407928
├── Ignored: 10998225 (88.64%)
└── Added to Cart: 1409703 (11.36%)
    ├── Purchased: 682334 (48.40%)
    └── Not Purchased: 727369 (51.60%)
```

Results
- Found smartphones had the highest purchase rate
- Items added to cart from 9-12 AM were most likely to be purchased
- The more expensive items had higher purchase rates
- Most items bought were added to cart <15m earlier.
- Found abandoned carts had higher viewership and more unique items interacted with



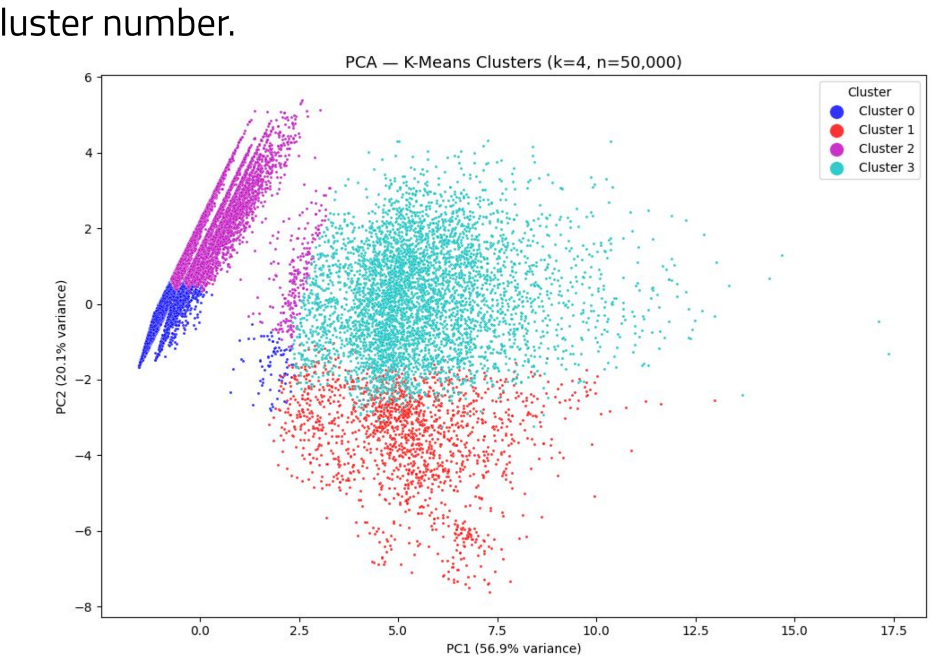
	price_bin	purchase_rate
0	<$10	0.324039
1	10–30	0.519488
2	30-100	0.502150
3	100–500	0.989236
4	$500+	1.142608

Feature Engineering

Four feature tables produced from the cleaned dataset:
- **user_features** aggregates event logs into one row per user. Captures RFM + additional behavioral metrics.
- **session_features** summarizes each session with event volume, time metrics, outcome flags, and revenue.
- **user_category_affinity** records weighted interaction scores between each user and category.
- **product_price_history** tracks daily median prices per product along with 30-day rolling statistics and price percentiles.

User Segmentation

Methods
- Performed K-means clustering using user_feature data (RFM + behavioral metrics)
  - Numeric features selected then normalized and standardized
- K-means performed for k=2 to k=10 on a sample of 500,000
- Optimal k of 4 selected using the elbow method and silhouette score, and K-means was applied to the entire user-base with this cluster number.



Results
- Four distinct clusters were produced
- By comparing median feature values, clusters were profiled and given the following labels:
  - Passive Browsers, Active Browsers, One Time Buyers, Engaged Buyers

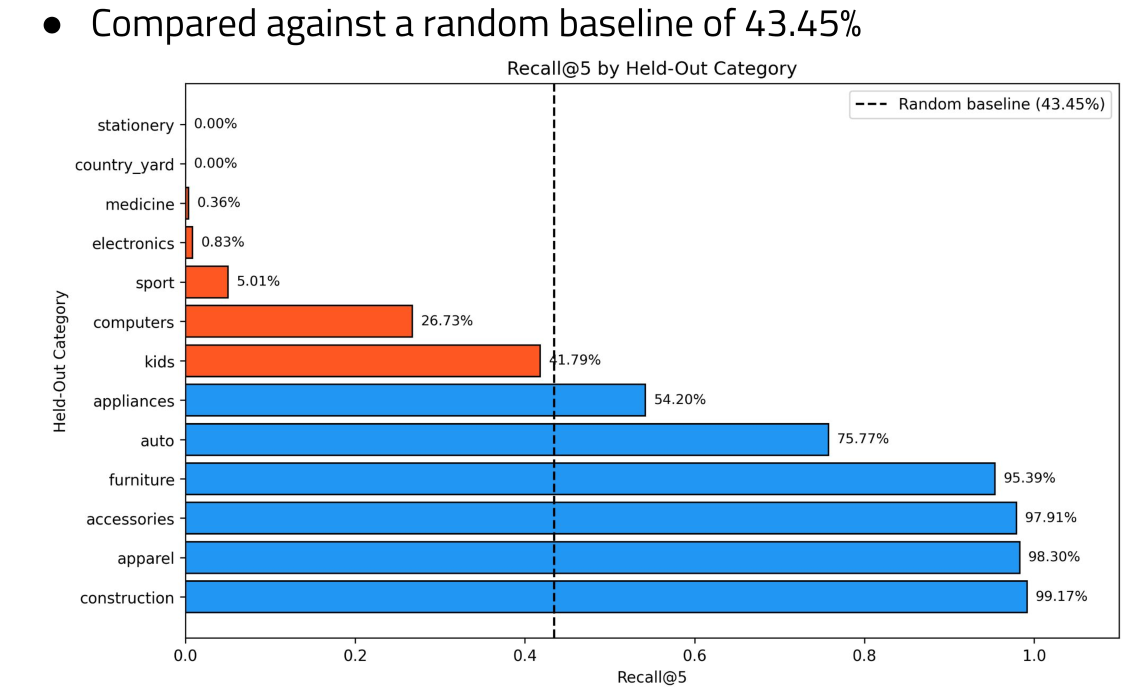
A/B Testing on User Segments

Methods
- Compared Active Browsers, Passive Browsers, One Time Buyers, and Engaged Buyers User Segments
- Used Mann-Whitney U Test to find statistically significant differences in the distributions of time spent browsing, money spent, and items viewed.
- Used 2-sample proportion Z test to compare the purchase and cart rates between all user segments

Results
- Found statistically significant differences between all clusters' distributions of time spent browsing, total product views, and money spent on site.
- Found statistically significant differences when it came to each cluster's true cart rate and purchase rate.

Collaborative Filtering

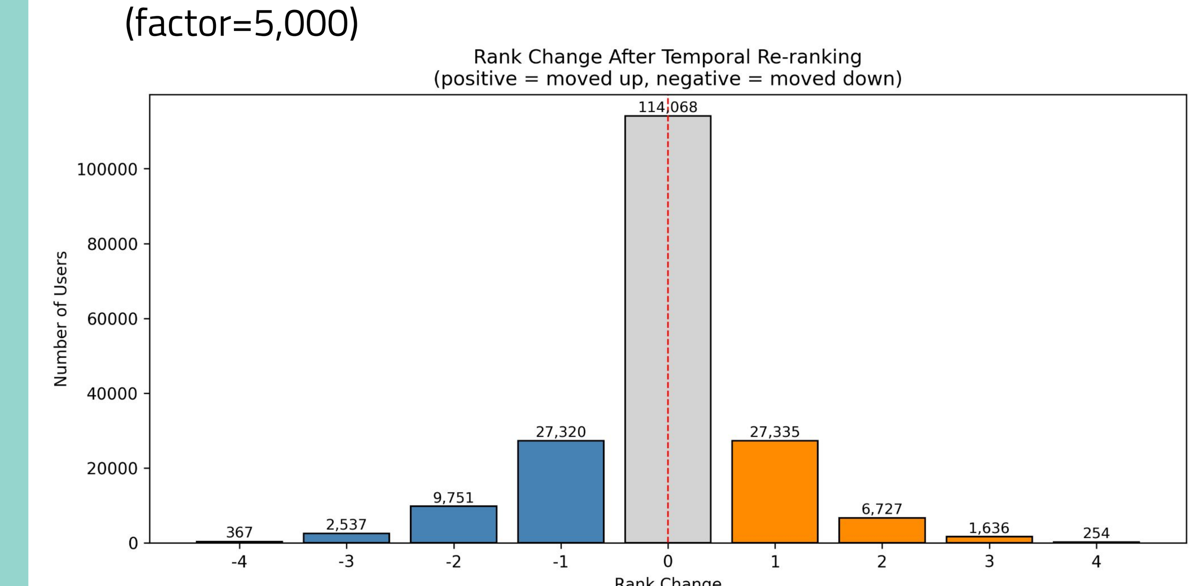
Methods
- Built a 13 x 13 category cosine similarity matrix from user-category affinity scores
- Applied shrinkage weighting (factor=50) to reduce unreliable low-support category pairs
- Leave-one-out evaluation: held out each user's highest affinity category as the test item
- Compared against a random baseline of 43.45%



Results
- Overall Recall@5: 33.50% - below random baseline
- Two failure modes: popularity bias in electronics (held out for 46.83% of users, 0.83% recall) and 3 sparse categories
- Excluding these four categories, Recall@5 rises to 62.39% - above random baseline
- Model performs well where sufficient interaction data exists

Time-Sensitive Recommendations

Methods
- Extended CF baseline as a post-processing re-ranking layer without building underlying model
- Category-level lift factors for hour and day; overall lift for seasonal dimension
- Lift factors combined by multiplication; shrinkage applied (factor=5,000)



Results

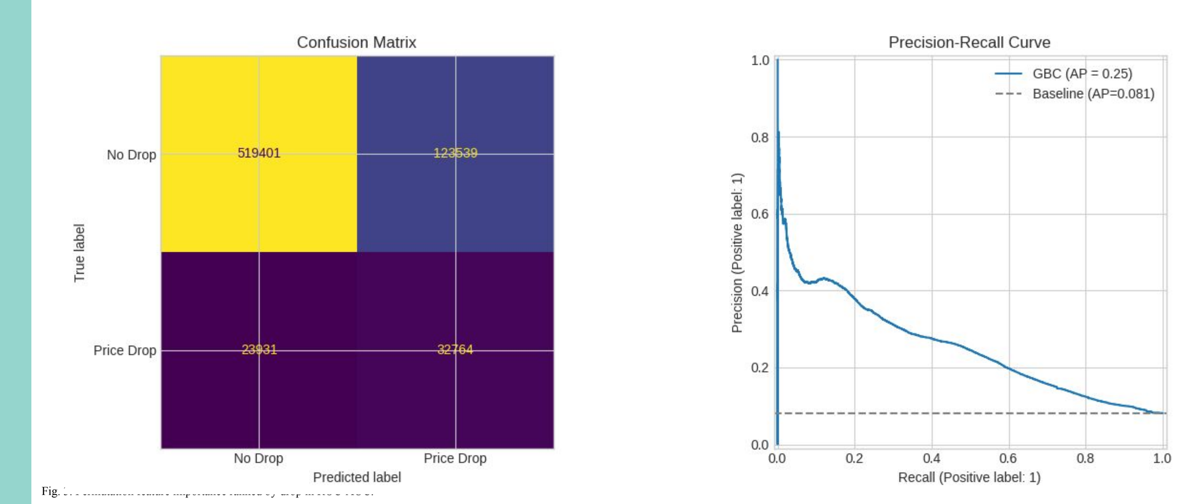
NDCG@5

Baseline (CF)	Temporal (CF + Re-ranking)
0.2117	0.2120

- Negligible NDCG@5 improvement (+0.13%)
- Near-symmetric rank changes - evaluation design not well aligned with temporal boosting
- True value requires live A/B test

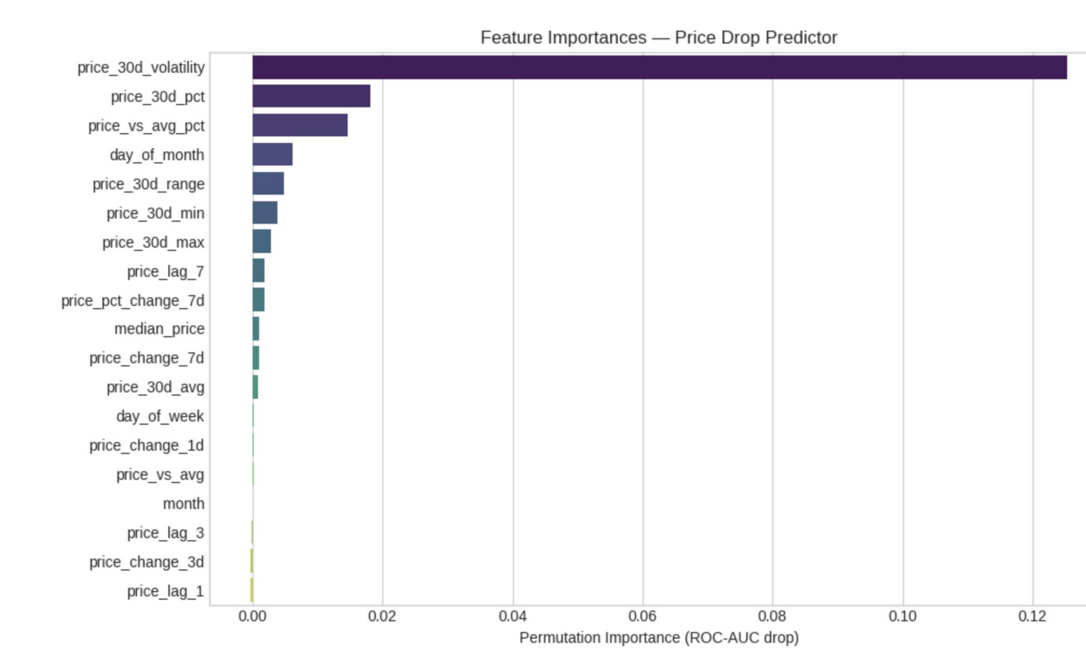
Price Timing Predictor

Methods
- Built on product_price_history (one row per product/date)
- 19 features: lag prices, rolling stats, volatility, distance from 30-day average, temporal
- Binary target: price drop >=5% in the next 7 days (7% positive, 93% negative)
- HistGradientBoostingClassifier with class_weight='balanced' to handle the 13:1 imbalance
- Temporal 80/20 train/test split to prevent look-ahead data leakage



Model correctly flagged 33,764 actual price drops

Results
- ROC-AUC: 0.744 vs 0.5 baseline
- Recall 0.832 at threshold 0.3
- Top feature: price_30d_volatility (importance 0.125)
- 150K products: 43% Buy Now, 37% Monitor, 19% Wait



Conclusion

This project provided hands-on experience applying core data science methods to a real-world dataset without a predefined project structure or guidance. Working on this project surfaced challenges rarely encountered in structured coursework, from memory management to evaluation design.

From an e-commerce perspective, the pipeline and results of this project demonstrates that raw behavioral data contains meaningful insights, but extracting those insights requires careful preprocessing, thoughtful modeling choices, and honest evaluation.